Monetary Commons: a new public institution for reframing Basic Income as a Personal Dividend

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Abstract

Departing from tax-funded UBI approaches this paper proposes a Personal Dividend that socialises the money-creation power hitherto monopolised by private banks. Central to the proposal is a voluntary migration of funds to the digital Monetary Commons, with increased personal dividends stabilising money supply through the negative money multiplier, thus reducing inflationary pressures, eliminating the need for Quantitative Easing, enabling revenue-neutral Pigouvian taxation, reducing wealth inequality, and enhancing systemic stability. Using agent-based simulation, we test adoption feasibility under various scenarios. The scheme consistently succeeds across diverse conditions, demonstrating its potential as a sustainable and voluntary alternative to conventional UBI frameworks.

Keywords
Universal Basic Income, Digital Monetary Commons, Money Multiplier, Personal Dividend

Section 1: Introduction

Arguments for a universal basic income have a long pedigree (Parijs, 2000; Standing, 2017) with eloquent advocates from both the right and the left of the political spectrum (see respectively Friedman, 2020; Standing, 2019) as well as detractors who consider it "a bad idea that never quite dies" (Giles, 2024). What unites advocates and detractors is their common – and, we argue, false – assumption that a universal basic income can only be funded through the fiscal route by a combination of taxes, government borrowing, and/or cutbacks in government expenditure, usually on existing social transfers (Calsamiglia & Flamand, 2019; Goldsmith, 2012; Marinescu, 2018; Nikiforos, Steinbaum, & Zezza, 2017).

In this paper, we argue for a radically different source of funding for a universal basic income that is located in the economy's monetary realm. We propose the creation of a Monetary Commons comprising a ledger and digital wallets for residents of the relevant jurisdiction managed by a public subsidiary of the central bank. Individual incentives (e.g., free transactions, offering the overnight central bank interest rate to depositors) encourage the migration of deposits from the private bank accounts of residents to their digital wallets. In so doing, the resulting negative money multiplier effect means that the monetary authorities not only can but *must* credit digital wallets with a Personal Dividend sufficient to maintain the quantity of money constant (to counter deflationary forces). Additionally, the Monetary Commons collects levies (instituted by lawmakers) on activities that damage the non-monetary commons (e.g., carbon taxes), as well as returns to socially produced capital, channelling these additional monies to everyone also through the Personal Dividend. In short, this paper proposes a Personal Dividend for everyone to be funded through an institution that is calibrated to socialise the money creation power hitherto monopolised by private banks. The proposed Personal Dividend is thus paid out from the universal pool of socially created value.

This innovation offers a clearer rationale for framing the Personal Dividend as a payment a person receives by right courtesy of being a co-owner of the jurisdiction's Monetary Commons. Besides the usual merits of the basic income for all (e.g., an income from caregivers and persons dependent on richer others, a substantial reduction in inequality and poverty rates, a significant rebalancing of social power with shared prosperity), a Monetary Commons-funded Personal Dividend has further transformative potential that includes the introduction of hitherto non-existent competition for the banks, less financial instability, and a new tool with which governments can impose credible, revenue-neutral, and highly progressive levies on companies and individuals damaging the commons.

Section 2 outlines the Monetary Commons and how its introduction and growth create monetary means for funding a Personal Dividend without increasing the money supply and without affecting the fiscal space with new taxes, spending cuts, or new debt. Section 3 presents the feasibility model simulating the migration of deposits from private banks to the monetary commons. Section 4 explores simulation designs and scenarios, while section 5 presents the simulation's results. Finally, section 6 discusses the results and conclusions.

Section 2: What is a monetary commons and how does it work?

Over 95% of all money in circulation in advanced economies is created by private bankers either directly or through the so-called money multiplier (Mankiw, 2010; Ryan-Collins, Greenham, Werner, & Jackson, 2014). One of the major contributors to the private banking sector's power to create money is its monopoly over the payment system – a monopoly strengthened by bank-dominated digital payment systems.

Here, we propose a public institution (e.g., the central bank or a subsidiary,) with a mandate to provide all eligible residents of the institution's jurisdiction with a free Digital Wallet, in the form of a banking app distributed through existing App Stores, into which participants can both transfer funds from their income or existing savings and make free digital payments or transfers to anyone else's Digital Wallet. To make this infrastructure of financial interest to the population, and for equity purposes, such Digital Wallets should accrue the central bank's overnight interest rate, which is higher than a commercial bank's deposit rate. As the use of Digital Wallets intensifies and more money migrates from private accounts held in commercial banks to the Monetary Commons, the marginal reserves shrink. Thus, the monetary base available to private banks decreases and so does the overall money supply.

To prevent the deflationary effects of a contracting money supply, the Monetary Commons would deposit a sum of monies into every Digital Wallet (i.e., the Personal Dividend) of a magnitude equivalent to the reduction in the money supply. A self-reinforcing cycle is thereby initiated: the more money migrates into the Digital Wallets, the more monetary space is created to grant everyone a dividend deriving from the strengthening of their Monetary Commons, the more attractive the Digital Wallets become and, thus, the more funds migrate from private banks to Digital Wallets, the more additional monetary space is created to increase the dividend; and so on. This dynamic process therefore funds dividends without straining the fiscal space by requiring new taxes or other types of fiscal injections.

The proposed Monetary Commons, in addition to opening up monetary space to fund a Personal Dividend in a non-inflationary manner, also enables the government to introduce revenue-neutral taxes on activities that harm other commons (e.g., carbon taxes). While such revenue-neutral taxes have been called for in the past (Monbiot, 2024), they have hitherto been notoriously difficult to administer in a manner that is credibly revenue neutral, the result being great resistance from a majority who are already bearing the burden of regressive indirect taxes. [1] However, when everyone has access to a Digital Wallet where regular dividends are paid to all, the task becomes both simpler and more credible.

More precisely, to offset the unfair burden on low-income households all the legislature needs do is specify that the taxes on activities that deplete our commons (e.g., carbon taxes) flow into the Monetary Commons and are fully re-distributed in the form of higher (flat) personal dividends. Similarly, dividends from a portion of Big Tech shares (Varoufakis, 2016) which represent a return on cloud capital that society has produced jointly but which Big Tech privatises, could boost the Monetary Commons further and provide a buffer for the social effects of increased automation (Chancel, Bothe, & Voituriez, 2024; Varoufakis, 2023, 2024). Since these taxes or levies do not flow into the Treasury but accumulate as unconditional income (or Personal Dividends) in everyone's Digital Wallets, they are revenue-neutral, highly progressive, and politically feasible – in that they ameliorate the suspicion if the public that the government retains a residual for other purposes. And as these taxes on 'bads' flow into the Monetary Commons, they help accelerate the migration of deposits from private banks to the Monetary Commons thus creating even more monetary space for higher Personal Dividends.

An additional macroprudential benefit from this dynamic is that, in deflationary times, the central bank has no need to resort to quantitative easing—a highly inefficient mechanism for dealing with deflation—since Digital Wallets make it possible to boost the money supply directly, equitably and with instant effect: simply by ratcheting up (in lump sum or variable rates) the Personal Dividend (see Supplementary Information A4).

Finally, the proposed Monetary Commons features several mechanisms to reduce private banks' influence in determining macroeconomic outcomes. The connection between routine individual customer services provided by banks and their involvement in speculative activities, which are often high-risk and difficult to regulate, has frequently been identified as a key factor in forming financial bubbles. This bundling implies that, during crises, bailing out the banking sector either directly or through practices such as

[1] For example, Marron and Toder (2014) note that "a carbon tax of 1.7 per cent of consumption imposes a burden of 2.1 per cent of consumption in the bottom decile, but only 1.3 percent in the top decile" and demonstrate that carbon tax is regressive regardless of the methodology used to calculate the burden.

Quantitative Easing, can seem the only viable response. The lack of any appreciable democratic oversight over these practices, and their significant role in contributing to financial instability and inordinate levels of inequality, underscores the need to separate the banks' socially responsible functions as intermediaries—such as facilitating transactions between savers and borrowers—from their higher-risk speculative operations. Under the proposed Monetary Commons the need for bank bailouts vanishes while the need for any Quantitative Easing is eradicated.

The question now becomes: What will it take for agents to shift their financial resources and transactions, in sufficient quantities, from their private bank accounts to their Digital Wallets so that the positive reinforcement dynamic described above takes hold? The simulation model of Section 3 addresses this question.

Section 3: Simulating the migration of money and transactions to the Monetary Commons

Politics and ideology will play a key role in the adoption of the Monetary Commons, in addition to the relative economic benefits of Digital Wallets. Transferring one's money from a private bank account to a Digital Wallet will depend on one's sentiment, ideology and trust in the new Monetary Commons. Questions about trust, cultivated antipathy to public institutions, social influence and mass psychology may prove as important as the individual financial benefits from the new Monetary Commons. As these dynamics can have no analytical solutions, simulations are one of the few quantitative guides to potential outcomes.

In contrast to many policy interventions whose success may depend on the effectiveness of implementation, external factors, or wider economic dynamics, the success of the Monetary Commons depends almost entirely on the sum of entirely voluntary individual decisions across a population. To understand the implications of the complex dynamics involved in the evolution of public opinion interacting with economic mechanisms, we developed an agent-based model of adoption of the Monetary Commons (MC). Given the impracticality of evaluating the possible socio-economic trajectories by means of large-scale experiments, this approach enables us i) to assess the feasibility of the scheme under different social and economic conditions including different implementation strategies, and ii) to identify factors that bear the most influence on its success as well as those that represent the highest risks.

3.1 Model Description

Consider a population of *N* agents who, at each time interval *t*, move a proportion of their funds to their personal Digital Wallet (DW) depending on their overall sentiment about the scheme, x. The sentiment is bounded in the range [-1 1] where -1 denotes full disagreement with the scheme and 1 full alignment. It is shaped by two motivating factors: social influence (opinion dynamics) and economic influence (perceived potential economic utility). We will look at each of them in turn.

3.1.1. Determining the sentiment

Social influence factor (*s*)

The sentiment-forming component of the model is based on the well-known opinion dynamics model and its recent implementation by Smaldino (2023). First, at each time interval t, each agent i, interacts with one other agent j, drawn randomly from among their first-degree neighbours (see below for the description of the social network structure). Their respective sentiments are updated symmetrically so that $x_i \leftrightarrow x_j$, as a function of bounded confidence (Deffuant, Neau, Amblard, & Weisbuch, 2000), which stipulates that the weight of influence, w_{ij}, is proportional to the distance between agents operationalised as the Manhattan distance, D_{ij} for each of their shared traits, k across the sentiment matrix of each agent (for our purposes, $k = 1$ denotes the sentiment towards the MC). The opinion dynamics is thus characterised by three features:

i) the influence is symmetrical, i.e., *when two agents meet they influence each other in equal measure*

ii) the influence weight is determined by the distance between agents' sentiments, i.e., we assume a threshold of influence such that *agents are more influenced by similar agents, and not influenced at all by agents that are dissimilar to them beyond a certain value*, and

iii) the strength of influence is inversely proportional to its distance from the mean, i.e., *the more extreme the opinion the less it changes as a result of social interaction*.

$$x_{is} \leftarrow x_i + \frac{1}{2} w_{ij} (x_{jk} - x_{ik})$$

$$x_{js} \leftarrow x_j + \frac{1}{2} w_{ij}(x_{ik} - x_{jk}) \tag{1}$$

$$w_{ij} = 1 - D_{ij} \tag{2}$$

$$D_{ij} = \frac{1}{K} \sum_{k=1}^{K} \left| x_{jk} - x_{ik} \right| \tag{3}$$

where *w* is the weight of influence determined by *D,* the difference between agents (*i,j*) based on *K* number of sentiments, *k.*

Economic influence factor (potential financial gain) (*e*)

The economic factor shapes the overall sentiment toward the Monetary Commons by considering the potential maximum financial gain from the scheme scaled by the individual's utility value of that gain and their perception of the risk involved. The interest rate differential, r^D*,* represents the difference in interest rates between the Monetary Commons and commercial banks – the presumption is that the former is the central bank's overnight rate and also that $f'(r^D) > 0, r^D$ lower bound to 0. The value of the Personal Dividend (PD), u, is scaled to its marginal value for each agent, i, depending on their income, Y_i.

$$x_{ie} \leftarrow T_i * (1 + r^D)u * \alpha Y_i^{\alpha - 1} \quad \text{, where} \tag{4}$$

T_i - individual's trust level expressing their perception of risk
α - utility of the financial gain
u - the value of the monthly Personal Dividend payment
r^D- interest rate differential between DW and commercial banks
Y_i - agent's monthly income

The agent's trust in the Monetary Commons, T_i, is a unique inherent characteristic of each agent denoting their trust in the monetary authorities (perhaps the state, more generally) so that any potential financial gains are scaled by the perceived probability [0,1] of obtaining them. One can conceptualise trust T_i as an individual's assessment of the probability that the financial gain will, in fact, materialise or, inversely, the risk, that is the probability that the scheme may be discontinued and the money in one's DW lost. In the current interaction of the model, T_i remains a constant throughout the run, that is, agents do not revise their perceived risk in light of the current situation – an assumption which will be relaxed in future iterations of the model. Finally, the economic factor is normalised to the [0,1] range to match the social influence component.

The overall sentiment of each agent is a sum of the social influence and economic factors weighted by the parameter β, that denotes the relative “importance” of each factor.

$$x_i \leftarrow (\beta x_{is} + (1 - \beta)x_{ie})(1 - |x_i|)$$
$$x_j \leftarrow (\beta x_{js} + (1 - \beta)x_{je})(1 - |x_j|) \quad (5)$$

The weighting is applied to each factor as proportions that add up to unity. In this iteration of the model, it is uniformly distributed in the population but future work should model the impact of a heterogenous distribution among the agents informed in empirical observations.

3.1.2 Determining the movements/migration of money

In each time interval, for amount of money in each individual’s digital wallet, DW_i, is determined by:

$$DW_i = u + x_i u \quad for \ -1 \leq x_i \leq 0)$$
$$DW_i = u + x_i Y_i \quad for \ x_i > 1 \quad (6)$$

This asymmetrical function of money movement mirrors the perceived aversion to or alignment with the scheme. That is, agents with the lowest sentiment do not interact with their DWs, but as the sentiment increases, they withdraw an increasing proportion of the funds. Agents with a neutral sentiment ($x_i = 0$) limit their activity to the withdrawal of the entirety of their personal dividend, while those with a positive sentiment ($x_i > 0$) also transfer other funds, here operationalised as their income, to DW. The total amount of funds in the Monetary Commons, MC, is defined by the money multiplier.

$$MC = 2/(1 + \rho) * DW_{Total} \quad (7)$$

The availability of funds in the Monetary Commons is assessed every twelve time steps (months), and if sufficient, the global value of personal dividend payments increases by a set amount. The value of personal dividend is capped at median income.

Section 4: Simulation design and scenarios

4.1 Experiment Design

4.1.1 Initialisation

The 'world' of persons is initialised as a network of 500 individual agents. The number of connections each agent has, i.e., the degree, is determined by parameter d. The network is modelled as a small world network, a lattice with a small number of random connections ($p = 8\%$). This structure is characterized by short path length but strong clustering, a set of characteristics common among social systems networks (Jin, 2001; Watts, 2004; Watts & Strogatz, 1998). The implementation of the small-world network follows (Smaldino, 2023) as follows.

Each agent is initialised with a personal level of trust, T_i, drawn at random from the range [0,1], and the monthly income is derived from the distribution of personal incomes based on the 2022 US census data (US Census Bureau, 2023) and capped at $400.000, thus representing 99% of US adult population. The annual income thresholds were cast into monthly incomes by dividing by 14 to account for the mean US income tax of approximately 15% (York, 2024). Although this simplification leaves out of our modelling issues related to the changes in the income distribution, it preserves the general shape of the income distribution. Modelling the full complexity of individuals' financial circumstances would require us to account for differing tax regimes across each state, the location-specific cost of living, and other personal circumstances, such as caring responsibilities, prior financial commitments such as mortgages or student loans, or the non-income related wealth of individuals including their inherited assets – important issues which are beyond the scope of this model. Note too that in the simulations reported below both the trust value and the income value are assumed to remain constant.

In every simulation run, each agent starts with an initial sentiment x, drawn randomly from the subset [-1, 0.5] of the possible range of permitted sentiments [-1, 1] following a conservative assumption that the initial range of opinions is likely to oscillate between hostility (-1) and moderate acceptance (0.5). The initial level of Personal Dividend and the maximum annual increase were set at a modest value of $100.

4.1.2 Scenarios tested

The model was run across four parameter axes generating 90 individual scenarios.

- **Polarised *vs* Consensus Society**

All scenarios were run under one of two social regimes: a polarised society and a consensus society. The dimensionality of the sentiment matrix, k, determines the overall trend towards polarisation (for low dimensions of the matrix) or consensus (for a high number of dimensions) in the baseline opinion dynamic model where economic considerations are absent; here, we tested the two extremes using two values: 3 dimensions for polarised and 7 dimensions for consensus societies. [See Smaldino, (2023) for the full description of the opinion dynamics model and its equilibrium states.]

- **Densely Connected *vs* Sparsely Connected Society**

The density of social connections, captured by the degree of nodes d, determines the potential speed of diffusion of the sentiment and the maximum potential diversity of interaction. Here, we tested low ($d = 2$), medium ($d = 5$), and high ($d = 10$) connectivity, all of which are in line with the observed range of close relationships held by individuals across different cultural contexts (Dunbar & Spoors, 1995)

- **Socially-driven *vs* Economically-driven Society**

The weighting between the social influence and the economic influence, β, denotes what proportion of the sentiment is shaped by each of the factors, where β determines the social influence and $1 - \beta$ the economic influence. β values were initialised across the full range [0, 0.25, 0.5, 0.75, 1.0], where 0 equates to the standard opinion dynamics model, i.e., 100% social influence, and 1.0 to a fully instrumentally rational utility maximising economic model corresponding to 100% economic influence.

- **Small *vs* Large Interest Rate Differential**

One of the key features of the Monetary Commons model is that the DWs accrue the standard overnight interest rate of the central bank. It is unlikely that any private bank will be able to match this interest rate long-term, thus, the interest rate differential, r^D, can be treated as a proxy for the level of attractiveness of the DW as a savings vehicle. Here, we tested three values of 1%, 4%, and 10%.

Table 1 provides an overview of the full parameter space. For each run, for each agent, the value of MC sentiment, x_i, was recorded in 12 time step intervals, together with the following measures:

- Overall (mean) **sentiment** towards the Monetary Commons

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N} \tag{8}$$

- **Polarisation**, operationalised as the variance among the pairwise distances between agents (see Smaldino, 2023, 2024)

$$P = var(D_{ij}) \tag{9}$$

- **Extremism,** operationalised as the proportion of agents with the sentiment above the absolute value of 0.9 (i.e., 10% at the extremes of the range) (see Smaldino, 2023, 2024)

$$E = \frac{\sum_{i=1}^{N} \left| x_i > 0.9 \right|}{N} \tag{10}$$

- Total amount of **money in the Monetary Commons** available for personal dividend payments, that is, the function of the money multiplier and the sum of money deposited on DWs.

$$MC = 2/(1 + \rho) * \sum_{i=1}^{N} DW \tag{11}$$

The length of the simulated time is 25 years. Each run was repeated 100 times to characterise fully the trends across the model's stochastic variance.

	Code name	Description	Values Range
N	num-agents	Number of agents	[2, ∞] **500**
	Opinion matrix	Agent's matrix of sentiments, x, regarding number, k of opinions. Randomly assigned at initialisation.	K * [-1 1]
K	num-opinions	The dimensionality of the sentiment matrix, low dimensionality converges towards polarisation, high dimensionality converges towards consensus	[1, ∞] **3 (polarised)** **7 (consensus)**
	initial-sentiment	The initial breadth of distribution of sentiment, x_i across the population	[-1 1] **[-1 0.5]**
	PD-change	The value of personal dividend is dynamically updated every 12 months if enough funds in the MC	static, **dynamic**
	opinion-strength	The weighting between social influence and economic gain factors	[0 1] **0, 0.25, 0.5, 0.75, 1**
r^D	int-diff	The difference between interest rates in DW and commercial banks	[0 1] **0.01, 0.04, 0.1**
u	PD-initial	if PD-change == dynamic, the initial value of personal dividend payments; if PD-change == static, the value of monthly personal dividend payments	[0, ∞] **100**
	network	Type of network	square lattice, ring lattice, **small world**
	spatial-interactions?	If network == square lattice, agents interact in von Neuman neighbourhood, otherwise, they choose a partner at random from the full population	[0, 1]
d	degree	If network == ring lattice or small world, the number of connections each agent has	[1, ∞] **2, 5, 10**
p	ntw-smallness	If network == small world, the proportion of links that are rewired.	[0 1] **0.08**
	income	Agent's monthly income. Randomly assigned at initialisation from a distribution of 2022 US incomes and capped at $ 400,000.	1 - 28 571
	trust	Agent's trust towards the MC. Randomly assigned at initialisation.	[0 1]
	mutility	The exponent of the marginal utility function. Same for all agents.	[0 1] **0.7**

Table 1. *The model's parameter space, including values used in initialisation and the range of tested values (in bold).*

Section 5: Simulation results

5.1 Results

The overall results show that the adoption of the Monetary Commons scheme is successful under a wide range of scenarios and parameter values. Adoption trajectories across most of the simulated scenarios trend upwards, following a similar path of an S-shaped curve. A slow initial uptake is followed by an inflexion point that marks a rapid increase in the amount of money deposited on individual Digital Wallets, which then levels out as the population's positive sentiment saturates. This resembles the common s-shaped innovation uptake curve (Henrich, 2001) and may indicate that conformist cultural transmission plays a significant role in driving the dynamics (Smaldino, Aplin, & Farine, 2018).

The overall dominance of successful outcomes also comes as no surprise given that the range of the social influence factor goes from negative to positive (range [-1,1]), while the economic factor can only take positive values in the range [0,1] since the personal benefit of receiving a Personal Dividend is always positive from the economic point of view. Thus, given the model setup, the economic factor and the Personal Dividend are bound to have a stabilising effect on the system, pushing the upward adoption trajectory. However, what varies between scenarios is the rate of adoption and some of the details in the shape of the curve that reflect the impact of the four tested factors (see 4.2.2).

From a systems science perspective, small variations within the same order of magnitude may seem insignificant as they reflect minor fluctuations around equilibria. Yet, in the real world, these variations represent socio-political realities that can significantly influence the success or failure of a policy, making them highly relevant. We therefore focus on the strength of the economic benefit stabilising effect against the other dynamics captured in the model.

All results for individual parameters are presented against the baseline of the conservative middle values, i.e., medium interest rate differential of 4%, moderate connectivity [$d = 5$], balanced weighting between social and economic influences [$\beta = 0.5$] and the polarised society scenario [$K = 3$].

Polarised *vs* Consensus Society

The MC adoption trajectories of the polarised and consensus society scenarios are virtually identical, following the flattened s-shape described above. This shows that even a moderate economic benefit can overcome the natural polarisation tendency if the social influence weight does not exceed the weight of the individual economic benefit ($\beta = 0.5$).

Societal polarisation has a limited effect on the system's dynamics, marginally slowing the adoption of the MC (Fig. 1) so that the funds on DW reach the reference level of the total income on average two years later than for consensus society.

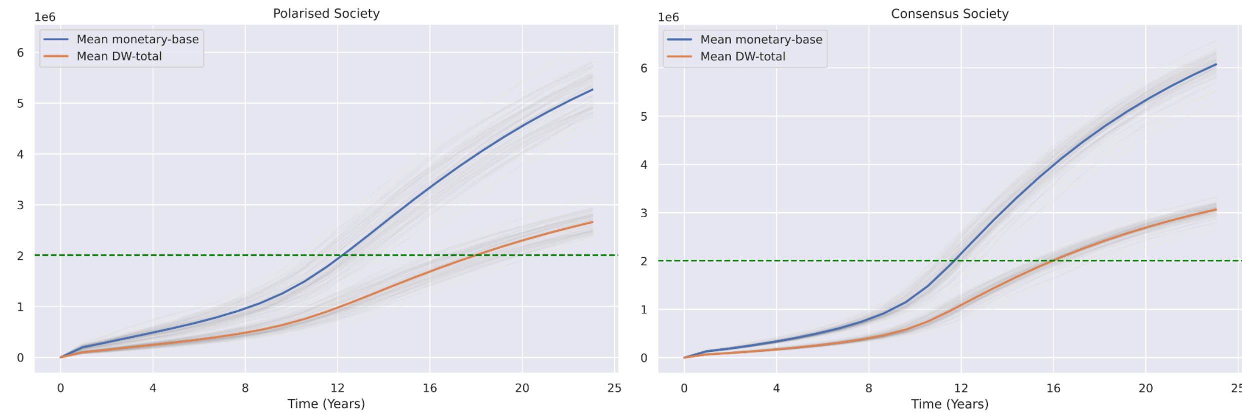

Figure 1. *Two trajectories of the MC adoption. Left: polarised society, right: consensus society. Grey lines show individual runs, colour lines mark the mean of all individual runs. The green reference line marks the sum of total income in the model (not including the PD payments).*

However, this economic trajectory has been reached through vastly different social dynamics (see Fig. 2). In the polarised society scenario, the level of polarisation continuously increased during the entirety of the simulated time. This was also matched by a parallel increase in extremism, meaning that more agents became set in extreme corners of the sentiment spectrum.

In the consensus scenarios, polarisation remained at a very low level of 3%, and extremism remained low, oscillating around 1% until some of the agents reached the high end of sentiment values around year 20. Thus in the polarisation scenario, those with the most positive sentiments counterbalanced those with negative views of the scheme, while in the consensus scenario, the bulk of the sentiment distribution gradually drifted upwards (Fig. 3). The overall economic effect kept its upward trajectory for both scenarios despite vastly different social dynamics behind it.

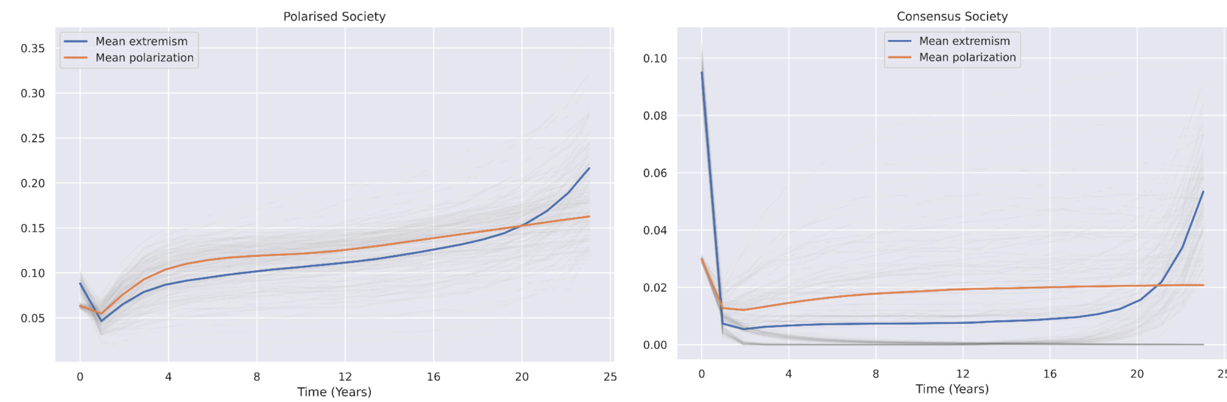

***Figure** 2. Social indicators. Left: polarised society, right: consensus society. Grey lines show individual runs, colour lines mark the mean of all individual runs.*

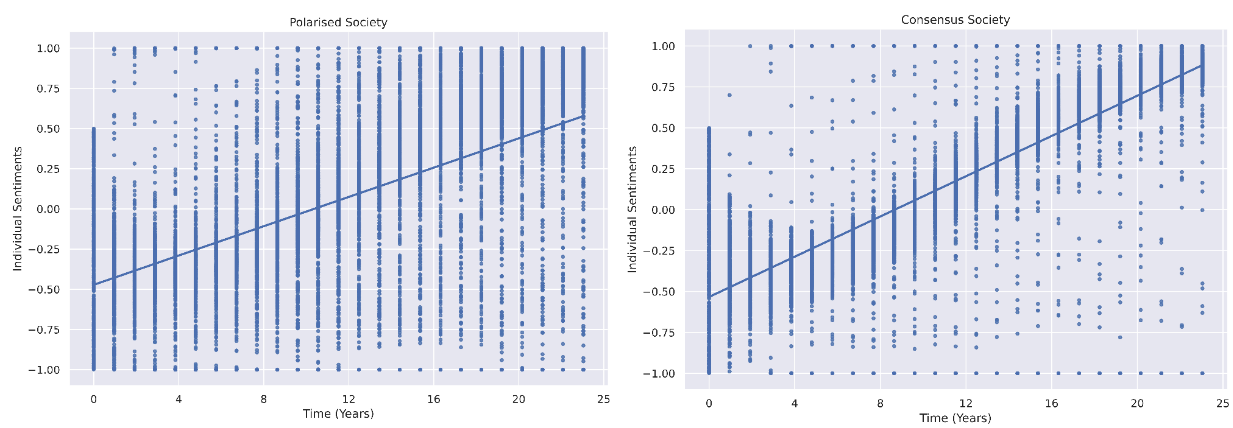

***Figure 3.** Example runs. Left: polarised society scenario, right: consensus society scenario. Each dot represents an individual sentiment towards the MC. Note the width of the distribution and the number of outliers in the extreme regions of the range for each scenario.*

Densely Connected *vs* Sparsely Connected Society

Despite the importance of information flows in shaping the dynamics of the model, the number of contacts each agent has (the degree of nodes d) has a small relative effect on the dynamics of the model (See Supplementary Materials C). Here, the small network structure already facilitates the spread of sentiments across the network so the degree does not make a substantial difference.

Small *vs* Large Interest Rate Differential

The interest rate differential has minimal impact on the model's dynamics (See Supplementary Materials C). This limited effect makes sense for two reasons. First, the interest rate only affects the initial personal dividend, which starts at a modest $100. Second, while a higher interest rate could theoretically incentivize individuals to move more income to their DW to maximize gains, this effect is dampened by diminishing marginal utility: as people earn more money, each additional dollar becomes less valuable to them. Thus, any increased earnings from higher interest rates would have proportionally less impact on individual decision-making. Even if the interest rate effect were stronger, it would only make the MC more economically attractive, further supporting our main conclusions.

Socially-driven vs. Economically-driven Society

The weighting between social and economic influence is a critical factor characterizing the dynamics of the model. As a reminder, parameter β specifies the proportion of sentiment influenced by opinion dynamics, with the remaining proportion driven by economic considerations, i.e., the potential financial gain of the individual. Specifically, when $\beta = 1$ (indicating 100% social influence), the model is fully governed by opinion dynamics. And when $\beta = 0$, the model represents a purely utility-maximizing economic framework.

Figure 4a shows how such purely utility-maximising agents fuel a rapid, almost linear growth of the scheme. However, even a limited (25%) social influence changes the shape of the curve, forcing a much slower initial flow of funds into the MC followed by a rapid “catch up” phase, when the scheme eventually takes off and reaches the reference level of funds within 15-20 years.

On the other hand, if social influence exceeds 50%, the uptake trajectory does not reach the acceleration phase of the s-shaped curve within the simulated time frame of 25 years (Fig. 4 d & e). Overall, it is only under the condition of 100% dependence on social influence that we see cases of failure of the MC scheme, i.e., trajectories that plateau or even fall. The implications of these findings are discussed further below.

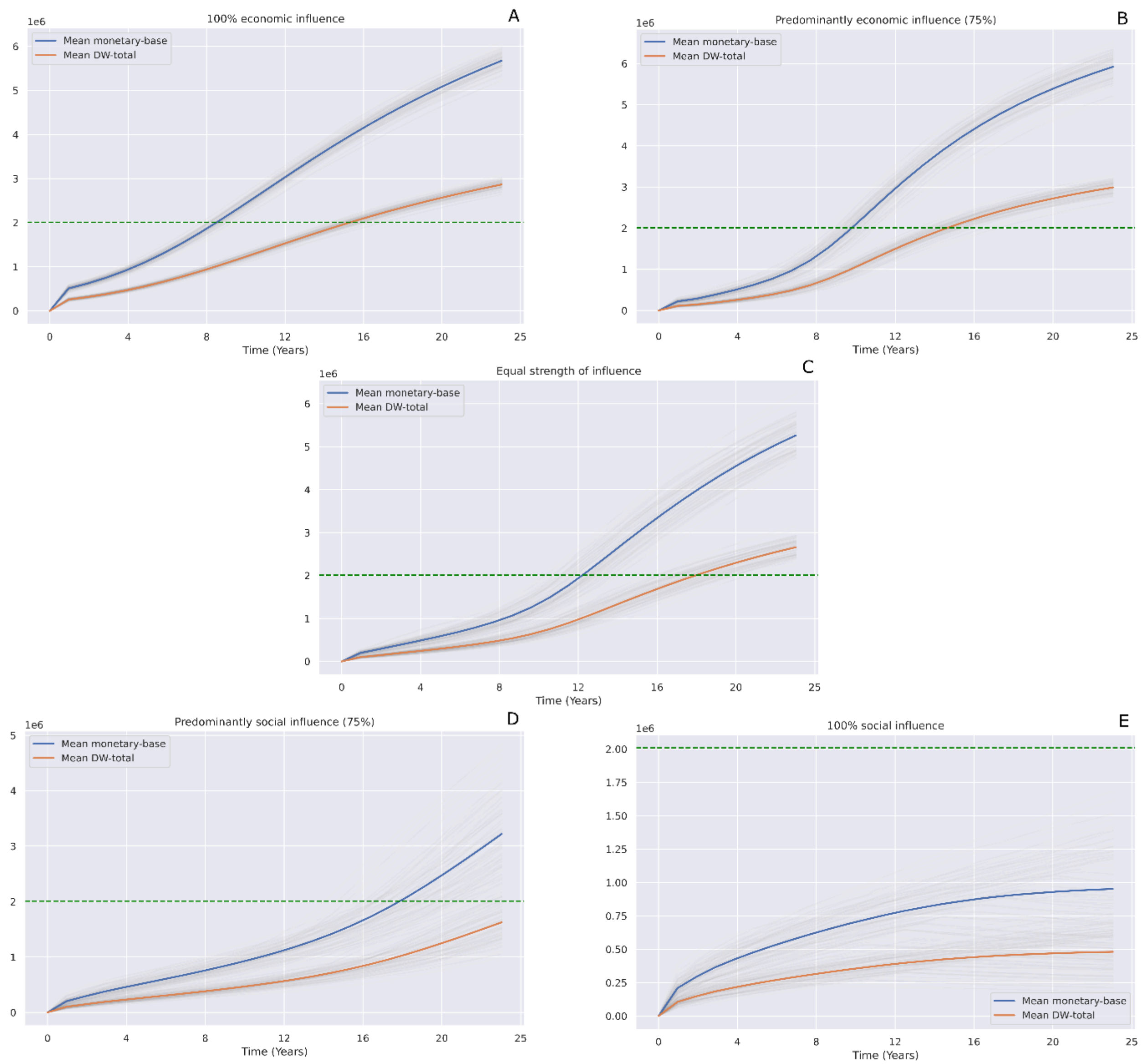

Figure 4. *The trajectories of MC adoption under different social vs economic influence scenarios.*

Despite the vastly different economic outcomes, the social indicators follow largely similar trajectories for polarisation and extremism, with the latter remaining at a low level of below 20% of the population for most of the time. In the economic-influence scenario, it is only after approximately 15 years that we see a sudden rise of extreme sentiments, in this case, these are entirely the maximum positive sentiments towards the MC that saturate the population (see Fig. 5).

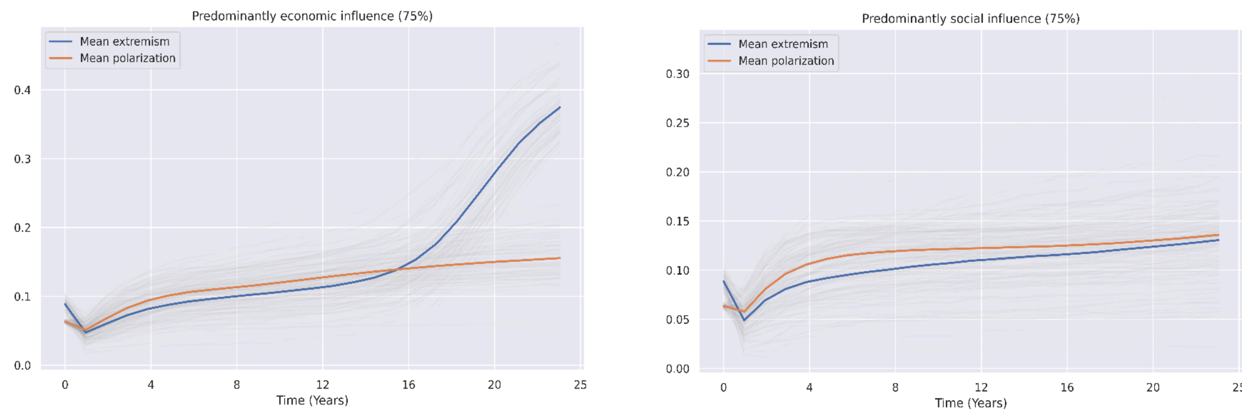

***Figure* 5**. *Social indicators. Left: predominantly economic influence (*$\beta = 0.25$*), right: predominantly social influence (*$\beta = 0.75$*).*

Section 6: Discussion

As an agent's participation in the Monetary Commons is entirely voluntary, its success depends on its adoption trajectory – a fundamentally social process. While these processes are notoriously difficult to model and predict, significant light has been shed on this process by adopting a foundational opinion dynamics model combined with an economic utility model for the purposes of assessing the likelihood that the Monetary Commons will be adopted.

The results demonstrate that the Monetary Commons is fully adopted under fairly general initial conditions, despite its voluntary nature and even in the presence of significant political resistance to its implementation. Under various conditions and scenarios, the sentiment shifts in favour of the Monetary Commons within the modelled time frame.

Nevertheless, it is worth noting that the shape of the adoption curve and, in particular, the slow initial "warm-up" phase at the predicted low initial uptake may impede adoption. Similarly, any short or medium-term pilot studies should consider a non-linear adoption rate which limits the representativeness of the data collected over the first few years.

That being said, the abstract nature of the model means that the time indicators (the 10-15 years inflexion point in the adoption curve) reflect several simplifying assumptions and should thus be treated with caution. Extensive calibration using real-world data should yield better temporal predictions regarding adoption rates.

One of the main findings of this study is that the same economic outcomes can arise despite different social processes underpinning them. Equally, vastly different economic outcomes can be reached without affecting social coherence. In the consensus society scenario, the mean sentiment towards Monetary Commons adoption moved toward the positive end of the spectrum, with most of the population oscillating close to the mean. But in the polarised society scenario, the sentiment trajectory, while similar, represented a middle point between two clusters of sentiments close to the extremes of the range and could therefore be considered less politically robust.

Reliance on social communication in individual decision-making emerges as the most significant factor driving the dynamics of the adoption process and the primary high-level risk to the scheme's success. Specifically, the model identifies the politicization of the scheme as the main risk because it can motivate agents to make decisions based mainly on socially constructed frames of reference, leaving out objective economic factors such as their financial gain. However, the results indicate that individual decisions do not need to be entirely driven by economic self-interest. Indeed, the adoption trajectories for populations (a) of fully utility-maximising agents and (b) of agents whose decisions are influenced equally by social and economic factors (50-50 balance of influence) are notably quite similar.

The presented model is designed to be abstract (often called “subjunctive” since it tests if-then scenarios – see David, Fachada, & Rosa, 2017). While this is appropriate for a feasibility study, future predictive models aiming to inform policy would require more robust data frameworks, including stronger validation. Thus, the model above should be considered as the first step in a series of iterations that build on with increasing complexity.

For example, the model only considers face-to-face communication and disregards other types of information flows (e.g., broadcast events). Similarly, it does not allow agents dynamically to update their “individual trust” depending on the scheme's success or dynamic changes to income structures that the scheme may trigger (both at the individual and societal levels). While these extensions are likely to improve the model’s predictive performance (e.g., vis-à-vis the rate of adoption), they are unlikely to change the general conclusion of this study regarding the feasibility of the Monetary Commons.

Section 7: Conclusion

This paper proposes a Monetary Commons that addresses two hitherto separate concerns: First, any universal basic income will be opposed by large majorities who refuse to pay higher taxes for it. Secondly, the present near monopoly of private banks over the payment system causes irrepressible financial instability and inexorable income inequality. A Monetary Commons, as outlined above, addresses both concerns.

As deposits migrate to the Monetary Commons, three are the repercussions: First, monetary space is created which can now fund (without new taxes and in a non-inflationary manner) a Personal Dividend for everyone. Secondly, banks lose their monopoly over transactions, thus giving banks powerful incentives to offer credit at more competitive rates as well as to restrict their rent-seeking, predatory behaviour. Thirdly, while financial stability is enhanced more broadly, the need for Quantitative Easing in deflationary times is eradicated.

Our modelling suggests that the general public will, under a broad range of plausible conditions, embrace the proposed Monetary Commons. While it demonstrates that the overall success of the scheme is plausible it also delineates the risks stemming from the fact that implementation hinges on institutional trust and patterns of collective behaviour.

The results highlight the critical role of political economy considerations, such as prioritising social acceptance, the reaction of vested interests (e.g., of the banking sector) and network effects over traditional economic incentives. A separate topic not touched upon in this study is the likelihood of a backlash by powerful vested interests against the idea of an, otherwise welfare enhancing, Monetary Commons.

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Data and code availability

The simulation model was developed in NetLogo 6.4.0 (Wilensky, 1999) and data analysis was performed using pandas 2.2.2 (The pandas development team, 2024). The model code, dataset of the results, and data analysis code can be found in this Zenodo record.

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Supplementary Materials A

On the Financial and Monetary Stability of the Monetary Commons

A1. The mainstream explanation of the money supply

The narrative dominating mainstream on how money is created is as follows:

I. The central bank mints central bank money and hands it over to private banks – this central money is thought of as ’real’ or base money (B).
II. As Jill saves some pennies, her savings are deposited in her bank account - i.e., they become deposits, say D, in her account with her private bank.
III. Then her bank lends to Jack a portion of Jill’s D equal to 1-ρ, where ρ is b’s *reserve ratio*, defined as the ratio of the overall funds that the bank must keep as reserves R; i.e. not lend out. So, if ρ=R/D=0.2, then the bank is not allowed to lend Jack more than 80% of Jill’s deposits.
IV. So, Jack receives (1-ρ)D and deposits it in his bank account. When he spends it on something Jack buys from, say, Kate, that sum of money ends up on Kate’s bank account
V. Kate’s bank (the same as Jack’s or another one), then lends (1-ρ)(1-ρ)D to, say, John … and so on and so forth.

To illustrate, suppose ρ=0.1 – i.e., banks keep 10% as reserves and lend 90% of their deposit. What happens to Jill’s $100 of savings/deposits? How does it grow? The following table explains how Jill’s $100 deposit/savings becomes $1000 of total loans (or total D) while the banking system continues to hold the original $100 as reserves (ρ).

BANK	Deposits (D)	Lending	Reserves kept (R)
Jill’s	100	90	10
Jack’s	90	81	9
Kate’s	81	72.9	8.1
John’s	72,9	65.61	7.29
…	…	…	..
Sums in the limit	D/ρ = $1000		D=$100

In this context, the central bank controls the money supply M by means of two control variables: (A) By adjusting the MONETARY BASE (i.e., how much B the central bank injects into the system), and (B) by deciding on the reserve ratio ρ, the monetary authorities (i.e., usually the central bank) can calibrate the money supply M to their desired level.

More precisely, $M = D+C$ and $B = R+C$, where C is the amount of money agents do not save (e.g., use as cash)

This can be re-written as:

$$\mu = (1+\lambda)/(\rho+\lambda)$$

where λ is the ratio of cash over bank deposits (C/D), and μ is the money multiplier; i.e., $M = \mu B$. E.g., if bank depositors want half of their deposits in cash ($\lambda=0.5$) and banks are obliged to keep 10% of deposits in reserve ($\rho=0.1$), then the money multiplier is 2.5 – i.e. for every dollar provided by the central bank, $2.5 are part of the economy's money supply. [Nb., Both λ and ρ are endogenous variables which means that the multiplier is also an endogenous variable]

A2. Three criticisms of the conventional theory of the money supply

<u>Criticism 1:</u> There is an assumption that banks need Jill's deposits in order to lend to Jack; and then Jack's deposits to lend to Kate etc. Banks conjure up deposits from thin air. Indeed, as long as they do not fall foul of the authorities' rules regarding their capital stock, they can fulfil the demand for loans if they want to.

<u>Criticism 2:</u> The mainstream theory assumes that the direction of causality is from reserves to loans; that banks need to know their reserves before they decide how much to lend. In reality, the direction of causality is the exact opposite: Banks first decide how much to lend and then look for the reserves to justify their lending levels. Who provides these reserves? Banks borrow from other banks and/or get the central bank to extend a credit line (i.e., increase their overdraft limit in their reserve account at the central bank). Indeed, there is zero econometric evidence that the monetary base B leads and the money supply follows. In fact, the opposite is more likely to be the case.

<u>Criticism 3:</u> There is no minimum reserve ratio in advanced countries like the US or the UK!

In summary, the idea that the central bank decides the money supply (through choosing B and ρ) is misleading. Most money is privately minted. Put differently, we live in a world of private money, with the bankers largely in control of its quantity.

A3. The Monetary Commons and Inflation: Why it is not inflationary

Under the Monetary Commons, every resident has a free digital wallet (DW) into which the Monetary Commons credits a fixed monthly personal Dividend, say u. If N is population size, each month the cost of the Personal Dividends equals U=Nu.

These Dividend payments come from three sources:

(a) LEVIES on commons depleting activities (e.g., carbon taxes) - L_c
(b) RETURNS, or dividends accruing to shares corporations have deposited with the Monetary Commons - V_c.
(c) New money generated by the Monetary Commons – B_c

In short, $U= L_c + V_c + B_c$

The two key features of the Monetary Commons are:

It operates outside the private money circuit of the private banks and, thus, each dollar in the Monetary Commons is just that: one dollar! In other words, there is no money multiplier within the Monetary Commons to turn that one dollar into $5, $10 or $100. This is the reason more money can be made available to increase the personal Dividend without increasing the overall money supply M: the key is the migration of dollars from our private bank accounts to our Digital Wallets. How does this work?

By encouraging us to transfer part or all of our bank deposits to our Digital Wallet,[i] the Monetary Commons motivates us to *sanitise* money that has already been created by the private banks. Suppose Jill (in the context of the example depicted in the Table of section 1) transfers $10 out of her $100 deposits in her private bank account to her Digital Wallet. A negative multiplier process commences. Given the banks' assumed reserve ratio of ρ=0.1, they will restrict the loans they have based on Jill's deposits (of $100) from $1000 to $900. In short, for every $10 that Jill shifts from her private bank to her Digital Wallet, total money supply shrinks by $90. Which means that the Monetary Commons can now expand its B_c level by $90 for every $10 that Jill shifts to her DW.

In summary, the Monetary Commons does not need to create new money to fund the personal Dividend. Simply by encouraging us (through offers of a higher interest rate on our deposits and no fees for our transactions) to shift existing monies from our private bank account to the Digital Wallet into which it pays our Personal Dividend, the Monetary Commons effects a net reduction in the money supply which, then, it compensates with upward adjustments to our Personal Dividend. Moreover, by receiving LEVIES (taxes on activities that damage the commons) and RETURNS (dividends in lieu of our collective contribution to capital accumulation), it sanitises that money too and, thus, creates even more monetary space from which to fund an even higher Personal Dividend.

A4. The Monetary Commons, Deflation and the eradication of the need for Quantitative Easing: How, unlike the current system, it is never susceptible to *liquidity traps*

Central banks can easily bring about deflation. By reducing the monetary base B, they shrink deposits (D) thus triggering the negative money multiplier effect (see the table in section A1 above) to suppress the money supply M. The result is a liquidity crisis that depresses prices and, ultimately, economic activity (including employment). However, the opposite is not true: When the economy is caught up in deflation or stagnation (recall the 2008-2022 period), central banks struggle to contain deflation (i.e., they often fail to create even a modest inflation; e.g., an inflation rate close to their usual target of 2%).

Example: Between the Great Financial Crisis of 2007/8 and 2022, we witnessed the total failure of central banks to create inflation. Inflation was stuck well below the Fed's and the ECB's 2% target and, despite energetic attempts to increase the monetary base B through so called Quantitative Easing, central banks were constantly failing to raise M, the money supply. Why? The reason lies within the Private Money system.

Quantitative Easing (QE) involves central banks creating reserves on a large scale, over a period of time, which they hand over to private banks in exchange for financial assets (e.g., bonds or mortgages) owned by the banks. The idea is that these purchases will push down interest rates on these bonds and mortgages, encouraging the financial sector to move away from investing in government bonds and instead lend to riskier ventures (through the so-called portfolio rebalancing effect). Such money creation is thus directed at the financial sector, using assets as collateral on the balance sheet of the central bank.[2]

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[2] The key assumption is that these operations are temporary: as soon as deflation is defeated and the economy is strengthened, the idea is that CBs would either let the bonds mature or sell them back to the private sector (mostly the same banks).

Alas, these operations do not work as planned. In sluggish or deflationary circumstances, banks do not lend the reserves they get from the central banks to small businesses or households because they fear that these ‘little’ people will not be able to repay the loans due to the sluggish economy. Moreover, if they do lend these reserves they lend them to conglomerates that use them to buy back their own shares (as opposed to spending it in new capital or to employ workers).

This is what John Maynard Keynes once described as a *liquidity trap*: a situation where central banks inject money into the monetary system but the money multiplier is close to zero – thus, extinguishing the central bank’s firepower. There is nothing that QE can do to help the economy escape from such a liquidity trap. All that happens in the end is asset price inflation, which boosts inequality but does not help escape the economy from price deflation’s gravitational pull.

In contrast, under the Monetary Commons, such a liquidity trap is impossible. All the central bank needs to do to boost the money supply is to increase the Personal Dividend!

A5. Conclusion

In terms of monetary and financial stability, the Monetary Commons offers two distinct macroeconomic advantages:

First, it allows for a Personal Dividend to be paid to everyone without a cent of new income or sales taxes or any inflationary boosts in the money supply – and it accomplishes this simply by pressing into public service the hitherto private money tree. Technically, this is achieved by taking advantage of the negative money multiplier so as to create monetary space for funding a Personal Dividend for everyone.

Secondly, in juxtaposition to QE that failed spectacularly to deal with liquidity traps and which caused massive increases in inequality, the Monetary Commons does exactly the opposite: it shrinks inequality, annihilates the exorbitant power of bankers, returns to society parts of the returns to capital and, crucially, offers everyone a trust fund by right.

When the idea of a Personal Dividend within a Monetary Commons reaches the ears of people in power, they will instantly try to terrorise you with horror stories of new taxes and galloping inflation. We now know how to debunk their fearmongering and to read between their toxic lines their fear of losing their monopoly of society’s money tree.

Supplementary Materials B

ODD Monetary Commons Adoption Model

Purpose

The Monetary Commons Adoption (MCA) model is a feasibility study designed to explore the potential trajectories of adoption or rejection of the Monetary Commons (MC) scheme which uses a digital banking system (Digital Wallet - DW) to deposit a Personal Dividend (PD) for each member. It models individual sentiment forming and decision making processes and translates them into population level consequences namely the adoption or rejection of MC. It identifies key factors influencing the rate of adoption. Given the impracticality of evaluating the possible socio-economic trajectories by means of large-scale experiments, using a modelling approach enables us i) to assess the feasibility of the scheme under different social and economic conditions including testing different implementation strategies and ii) to identify factors that bear the most influence on its success as well as those that represent the highest risks. The overarching goal is thus to evaluate the viability of the MC framework and understand how individual factors such as social influence, economic motivation or the structure of their social network shape individuals' interactions with the framework and the probability of adopting it.

Entities, State Variables, and Scales

The MCA model comprises a population of agents connected with each other in a small-world network. There is no explicit spatial component other than the node's network position with respect to other nodes. The following state variables characterize each agent:

- **Opinions** A vector of individual opinions, with the first value representing the agent's sentiment towards the MC framework. Sentiment ranges from -1 (maximum negative) to 1 (maximum positive). The sentiment towards the MC framework determines the agent's interactions with the Digital Wallet (DW) (see submodel *Economic Dynamics).*
- **Income** A static attribute representing the agent's monthly income. It does not change over the course of the simulation. The distribution of incomes across the population is based on USA 2022 census data (US Census Bureau, 2023).
- **Digital Wallet (DW)** The amount of money held in the agent's DW, where funds can be added or withdrawn based on the agent's sentiment towards the MC framework.
- **Trust** An individual characteristic of each agent reflecting their perception of risk associated with the MC scheme, conceptualised as the risk of a default by the MC. It is inherent to each agent and unchangeable over the course of a run.

The simulation operates over a 25-year period, represented by 300 timesteps, with each step corresponding to a month.

Process Overview and Scheduling

In each timestep, the model goes through two processes.

1. **Sentiment Formation** Agents form their sentiment towards the MC framework based on two intertwined processes: social influence and economic gain evaluation. Social influence is modelled using Smaldino's (2023) opinion dynamics, while the economic gain is assessed through the potential financial benefits an agent perceives from the scheme, adjusted for their income and individual level of trust towards state intervention.

2. **Interaction with Digital Wallets** Agents interact with their DWs based on their sentiment. Negative sentiments lead to limited engagement with the scheme consisting predominantly of the withdrawal of funds, while positive sentiments result in additional funds being added to the DW. Specifically, agents with the maximum negative sentiment (value: -1) refrain from any interaction with DW meaning that they do not alter their DW funds. As the sentiment value rises agents show increasing engagement resulting in an increasing proportion of funds being withdrawn. This process peaks at the neutral sentiment (value: 0) so that an agent with a neutral sentiment will withdraw their entire Personal Dividend (PD). A positive sentiment indicating trust in the MC translates with an increase in the everyday use of DW. Agents with positive sentiments (value between 0-1) add funds from their income pool to DW proportional to their sentiment towards the scheme. Thus, an agent with the maximum positive sentiment (value: 1) will keep their PD and additionally transfer their entire income to the DW.

In addition, every 12 timesteps, that is, annually, the feasibility of increasing the value of the Personal Dividend (PD) is evaluated based on the availability of funds in the MC, up to a ceiling set by the median income level.

The model's key output measures collected at each time step include the sum of all agents' DWs, the total aggregated funds within the Monetary Commons which are the function of funds available on DWs and the money multiplier as well as social indicators, namely extremism and polarization. Adapted from Smaldino's 2024 opinion dynamics model, extremism is conceptualised as the proportion of sentiments falling within the 10% most extreme values (either positive or negative). Polarisation is recorded as the variance of opinions across the full opinions vector.

Design Concepts

Basic Principles
The model integrates two classic models albeit from two disciplines, a foundational social science opinion dynamics model and a simple economic model of individual incentives.

Emergence
The model has a narrow scope as a feasibility study and thus it does not capture emergence in its standard definition (Epstein, 1999). Nevertheless by operating across several scales of abstraction: the processes behind individual decision-making influenced by social and economic factors to the monetary economy of the population the model simulates how localized interactions can lead to broader systemic changes. It marries social factors such as the social determinants of adoption or resistance to an economic policy with economic modelling both at the level of the individual (personal incentives) and the system as a whole (the money multiplier). In addition, collective social patterns such as polarization or consensus emerge as a result of individual-level interactions. The model's objective was to assess the feasibility of the MC framework at a societal level but the socio-economic dynamics captured are of broader interest.

Adaptation
There is one adaptive process in the model: agents adapt their economic behaviour based on their evolving sentiment towards the MC which influences whether agents remove funds from their Digital Wallet or add to it. With the aim to isolate the key drivers of the observed dynamics several factors that could be modelled dynamically, i.e., adapt over the course of a run, were left static. In particular, the agent's income remains constant throughout the simulation as does their risk perception which does not change in response to the scheme's popularity or success. This constitutes conservative assumptions which, if relaxed, would further tip the dynamics towards faster adoption of the MC scheme.

Objectives
The model does not assign explicit long-term goals to agents; instead, it simulates how individual, short-term adaptation leads to a societal-wide outcome. Each agent maximizes personal utility conceptualised as consisting of two factors: 1. the socially-driven preference to align with the consensus of one's closest social contacts and 2. the individual economic utility from participating in the MC. Agents do not hold predetermined personal preferences towards or against the MC and their political, social or economic views are not modelled explicitly.

Sensing
Agents have access to variables determining their economic situation: their income, the funds in their DW, and their level of trust. At each time step, they interact with one of their network connections and can thus perceive the sentiments of one other agent. This sensing does not extend to an awareness of the evolving sentiments of the entire population. Agents are also not aware of the total amount of money available in the MC.

Interaction
Agents interact with their closest network connections. At each time step, agents randomly select an interaction partner and exchange opinions to a degree determined by the distance between their respective sentiments. Each agent interacts with their individual DW but has no access or direct influence of the MC.

Stochasticity
At initialisation, agents are randomly allocated: i) monthly income, ii) level of trust, iii) initial sentiment towards the MC, and iv) a position within the network. The choice of an interaction partner is also random.

Observation
Agents' sentiments are collected at each step, together with the following population-wide measures:

- Overall (mean) **sentiment** towards the Monetary Commons

  $$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

- **Polarisation**, operationalised as the variance among the pairwise distances between agents (after Smaldino, 2023)

  $$P = var(D_{ij})$$

- **Extremism,** operationalised as the proportion of agents with the sentiment above the absolute value of 0.9 (i.e., 10% at the extremes of the range) (after Smaldino, 2023)

  $$E = \frac{\sum_{i=1}^{N} \left|x_i > 0.9\right|}{N}$$

- The total amount of **money in the Monetary Commons** that is thus available for personal dividend payments, that is, the function of the money multiplier and the sum of money deposited on DWs.

  $$MC = 2/(1 + \rho) * \sum_{i=1}^{N} DW$$

Initialisation

Global parameters
All experiments have been performed with a population of 500 agents distributed on a small world network (see submodel *Network Formation*), i.e., a fully connected ring lattice with 8% of links randomly rewired.
The initial value of the Personal Dividend (PD) was set at 100 USD. The ceiling for the annual increases of PD was set at the median income of the population.
The simulation runs for 300 time steps, equivalent to 25 years.

Agent parameters
Each agent was seeded with a random sentiment value in the range [-1 0.5], that is, going from max negative to moderately positive. Here the conservative assumption was that any new intervention is likely to be met with initial scepticism. Table S1 provides the values of all parameters at initialisation.

	Code name	Description	Values
N	num-agents	Number of agents	[2, ∞] **500**
x_{ik}	Opinion matrix	Agent's matrix of sentiments, x, regarding number, k of opinions. Randomly assigned at initialisation.	K * [-1 1]
K	num-opinions	The dimensionality of the sentiment matrix, low dimensionality converges towards polarisation, high dimensionality converges towards consensus	[1, ∞] **3 (polarised)** **7 (consensus)**
	initial-sentiment	The initial breadth of distribution of sentiment, x_i across the population	[-1 1] **[-1 0.5]**
	PD-change	The value of personal dividend is dynamically updated every 12 months if enough funds in the MC	static, **dynamic**
β	opinion-strengt h	The weighting between social influence and economic gain factors	[0 1] **0, 0.25, 0.5, 0.75, 1**
r^D	int-diff	The difference between interest rates in DW and commercial banks	[0 1] **0.01, 0.04, 0.1**
u	PD-initial	if PD-change == dynamic, the initial value of personal dividend payments; if PD-change == static, the value of monthly personal dividend payments	[0, ∞] **100**
	network	Type of network	square lattice, ring lattice, **small world**

	spatial-interactions?	If network == square lattice, agents interact in von Neuman neighbourhood, otherwise, they choose a partner at random from the full population	[0, 1]
d	degree	If network == ring lattice or small world, the number of connections each agent has	[1, ∞] **2, 5, 10**
p	ntw-smallness	If network == small world, the proportion of links that are rewired.	[0 1] **0.08**
Y_i	income	Agent's monthly income. Randomly assigned at initialisation from a distribution of 2022 US incomes and capped at $ 400,000.	1 - 28 571
T_i	trust	Agent's trust towards the MC. Randomly assigned at initialisation.	[0 1]
α	mutility	The exponent of the marginal utility function. Same for all agents.	[0 1] **0.7**

Table S1. *The model's parameter space, including values used in initialisation and the range of tested values (in bold).*

Input data

Given the nature of the model, most parameters have been abstracted, e.g., the population is held constant at 500 agents.
The distribution of agents' incomes is based on the US Census data for personal income in 2022 (US Census Bureau, 2023). Each agent's income is sampled from the overall income distribution binned in 1-25k, 25k-50k, 50k-100k, 100k-200k, 200k-400k intervals denoting annual income with a probability equal to the proportion of the population that falls into the given bin. To recalculate from annual to monthly income and account for income taxes, the value was divided by 14 which approximates imperfectly into the 2021 average income tax rate of approx. 15% (York, 2024). This approximation was deemed parsimonious in light of the overall level of abstraction of the model and the lack of detailed modelling of more important wealth determinants such as capital, assets or spending patters, dependants and other factors.

Submodels

1. Network Formation
The algorithms used to initialise the network are lifted directly from (Smaldino, 2023, 2024) The network structure is fixed throughout the simulation. The model comprises different network topologies:

- **Random connections**: Agents are connected to others at random with no spatial element (so-called “bag of agents”).
- **Spatial networks**: Agents are connected with neighbors in von Neuman neighbourhood.
- **Ring-lattice:** Agents are connected aspatially. The number of connections each agent has is determined by the *degree* parameter that controls the density of the network, ranging from sparse to highly connected.
- **Small-world**: Agents are connected as in the ring-lattice setup but a proportion of the links are rewired to a random node.

All experiments have been run on the small-world network with 8% of links rewired.

2. Social Motivation

Social opinion dynamics are modelled using the opinion dynamics framework based on (Smaldino, 2023, 2024). This interaction follows the following algorithm:

1. At each time step, a random agent, *i*, selects another agent (partner), *j*, within its network neighbours.
2. The distance, *D*, between their opinions is calculated as the mean difference across the vector of sentiments (opinions), the dimensions of which are determined by num-opinions parameter. This distance reflects how similar or dissimilar the two agents' opinion profiles are, with no special weight given to their sentiment towards the MC framework.
3. The influence of the interaction is weighted by the social similarity, *w*, between agents, defined as *1−distance*.
4. Sentiments, *x*, are updated based on the weighted influence of the interaction between agents, *i, j*:

$$x_{is} \leftarrow x_{i} + \frac{1}{2} w_{ij} (x_{jk} - x_{ik})$$

$$x_{js} \leftarrow x_{j} + \frac{1}{2} w_{ij} (x_{ik} - x_{jk})$$

$$w_{ij} = 1 - D_{ij}$$

$$D_{ij} = \frac{1}{K} \sum_{k=1}^{K} \left| x_{jk} - x_{ik} \right|$$, where

w - weight of influence determined by D
D - difference between agents (*i,j*) based on *K* number of sentiments, *k*

Thus, social opinion dynamics are characterized by three factors. First, the influence is symmetrical, i.e., when two agents meet they influence each other in equal measure. Second, since influence weight is determined by the distance between agents, this implies a threshold of influence, meaning that agents are more influenced by similar agents, and not influenced at all by agents that are dissimilar to them beyond a certain threshold. Third, the strength of influence is inversely proportional to its distance from the mean, i.e., the more extreme the opinion the less it changes as a result of social interaction.

3. Economic Motivation

Agents' perceived potential economic gain from the MC framework is a function of their income, the value of the Personal Dividend (PD), and is adjusted by their inherent risk perception (trust) in the system. While the social motivation varies between [-1 1], i.e., can be negative or positive, the economic gain is always positive due to the nature of the personal dividend, that is, the fact that it is an unconditional transfer of funds that does not incur any costs. This transfer is not dependent on the agent's income or sentiment towards the scheme (but see below for the *Economic Dynamic submodel*).

1. For each agent, the potential financial gain is computed as:
   $x_{ie} \leftarrow T_i * (1 + r^D)u * \alpha Y_i^{\alpha-1}$, where

   T_i - individual's trust level expressing their perception of risk
   α - utility of the financial gain
   u - the value of the monthly UBI payment
   r^D - interest rate differential between DW and commercial banks
   Y_i - agent's monthly income

2. The potential financial gain is then normalized using global minimum and maximum values for economic outcomes, ensuring that the value is compatible with the social motivation component of the sentiment.

Thus the marginal utility is modelled as a function of the agent's income and can be conceptualised as the value of PD money to an agent given how much funds are already available to them. The trust parameter determines the agent's perception of the risk associated with the scheme. It is initiated at random and remains unchanged throughout the simulation. It can be conceptualised as the agents' inherent belief or the lack thereof in the stability of governmental interventions, translating into their personal evaluation as to whether the MC will not default and they will be able to access the funds accumulated on their DW.

The final sentiment is calculated as a weighted sum of social influence and economic benefit:

$$x_i \leftarrow (\beta x_{is} + (1 - \beta)x_{ie})(1 - |x_i|)$$
$$x_j \leftarrow (\beta x_{js} + (1 - \beta)x_{je})(1 - |x_j|)$$

The **opinion-strength,** β, parameter controls the balance between social influence and economic considerations.

4. Economic Dynamics (Digital Wallet Interactions and Monetary Commons Funds)

Agents interact with their Digital Wallets (DW) based on their updated sentiments toward the MC. The amount of money each agent adds or withdraws from the DW is governed by an **asymmetrical mapping curve**, reflecting conservative assumptions about behavior:

- **Negative sentiments** (values below 0) Agents withdraw funds from their DWs. The more positive the sentiment, the more money they withdraw. Thus, agents at the maximum negative sentiment [-1] do not with their DWs at all.
- **Neutral sentiment** (value at 0) Agents withdraw all PD funds from their DWs.
- **Positive sentiments** (values at or above 0) Agents add funds to their DWs. Agents with a sentiment of 1 transfer their entire income, along with the PD, into the DW.

At each time step, the total amount of funds available within the MC is calculated as the sum of all agents' DW balances. The **money multiplier** (Mankiw, 2010) determines the total funds available within the MC.

$MC = 2/(1 + \rho) * DW_{Total}$

Each year, if the total funds in the MC exceed the minimum necessary to fund the increase, the value of the PD increases by 100 USD, up to the ceiling defined by the median income.

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Supplementary Materials C

The model code, dataset of the results, and data analysis code can be found in this [Zenodo record](#).